

ADVANCED MACHINE LEARNING · WORKSHEET 09

Bias, Variance and the Bias–Variance Tradeoff

Estimators · Underfitting & Overfitting · MSE Decomposition · Practical Diagnosis

Lecture 9

Name _____	Branch / Year _____	Date _____
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PART I: UNDERSTANDING BIAS AND VARIANCE**SECTION 1 Setting the Stage: $Y = f(x) + \varepsilon$** **▷ HOOK**

Suppose we want to model a student's **Stipend** which depends on three input features: CGPA (X_1), IQ (X_2), and Number of Projects (X_3).

There exists a **true hidden relationship** or function that determines the Stipend. But even then, the observed value will not be exactly equal to the true pattern, because there is always some randomness in the system.

We represent this randomness with random noise (ε):

$$\text{Stipend} = f(\text{CGPA}, \text{IQ}, \text{Projects}) + \varepsilon$$

Let $X_1 = \text{CGPA}$, $X_2 = \text{IQ}$, $X_3 = \text{Projects}$, $Y = \text{Stipend}$.

$$\Rightarrow Y = f(X_1, X_2, X_3) + \varepsilon$$

Let us just call the combination of input variables X_1, X_2, X_3 simply as X :

$$Y = f(X) + \varepsilon$$

where:

- $f(X) =$ _____ (the true hidden relationship)
- $Y =$ _____ (observed output)
- $\varepsilon =$ _____ (random noise = irreducible error)

In words:

$$\text{Observed Output} = \text{True Pattern} + \text{Random Noise}$$

Properties of the noise:

By definition, the mean of all random noise (white noise) is 0:

$$E(\varepsilon) = \underline{\hspace{2cm}}$$

Say ε has some variance σ^2 , where σ is the standard deviation of white noise:

$$\text{Var}(\varepsilon) = \sigma^2 = E(\varepsilon^2) - [E(\varepsilon)]^2 = E(\varepsilon^2) - 0$$

$$\Rightarrow \text{Var}(\varepsilon) = \sigma^2 = \underline{\hspace{2cm}}$$

What does white noise look like?

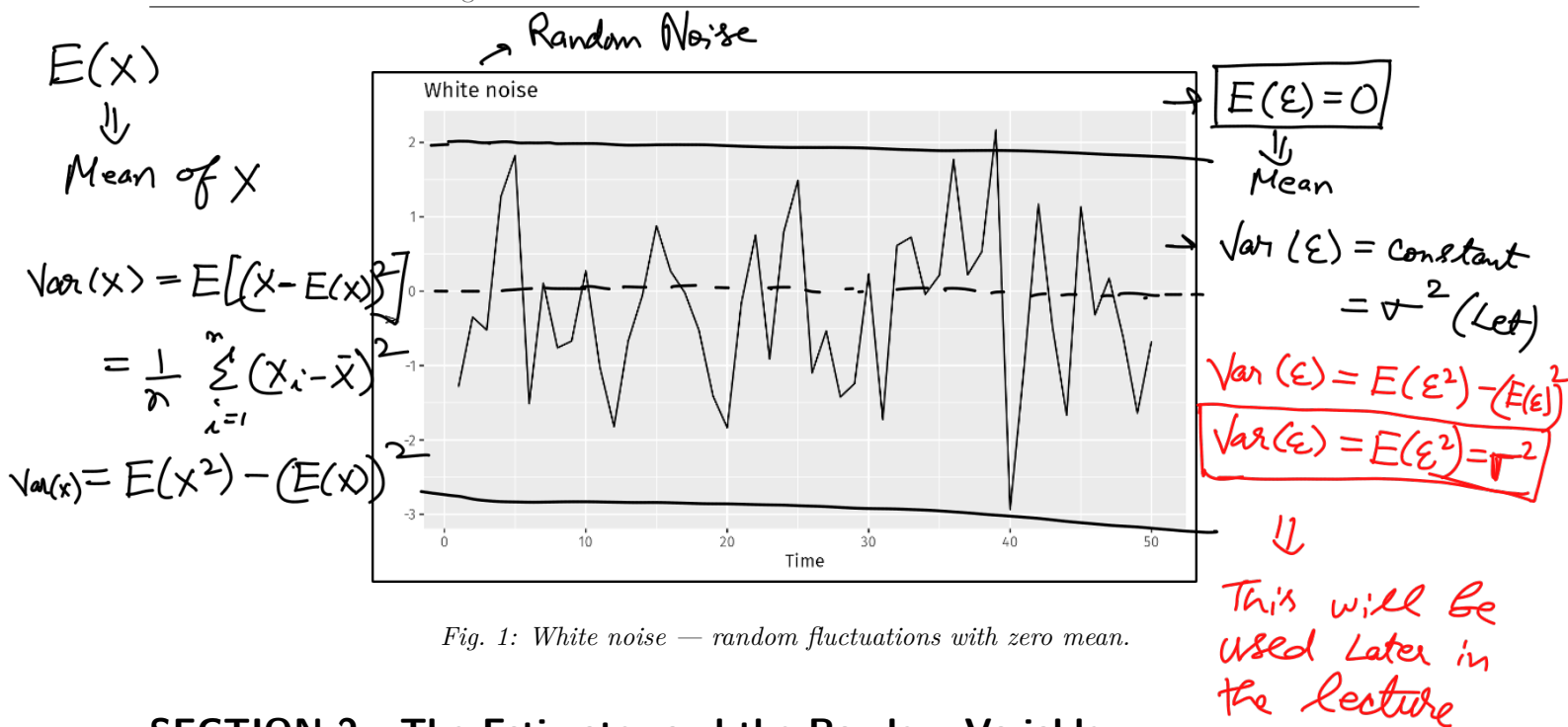


Fig. 1: White noise — random fluctuations with zero mean.

SECTION 2 The Estimator and the Random Variable

▷ HOOK

The true function $f(X)$ exists for the whole population of data. We don't know what this true function is. We can try to best estimate it using our **sample dataset** (which is derived from the population).

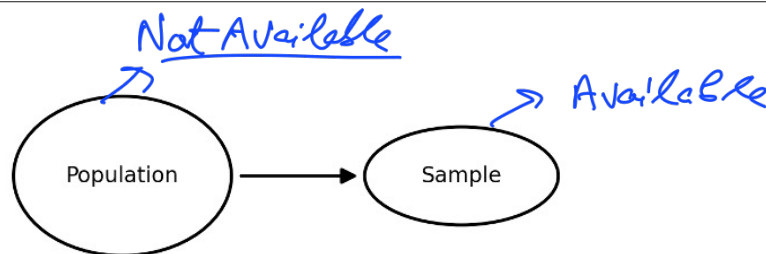


Fig. 2: We draw a sample from the larger population.

Using the training dataset, we build a model: $\hat{f}(x)$. This is called an **estimator** of the true function.

Examples of estimators:

1. Sample mean estimates _____
2. Linear Regression estimates _____
3. Decision tree estimates _____

Imagine collecting **different training datasets** from the same population:

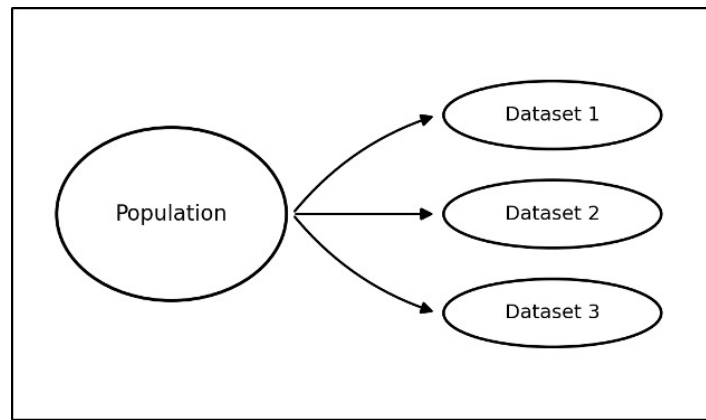


Fig. 3: Different sample datasets drawn from the same population.

★ KEY INSIGHT

- Sample Dataset 1 $\rightarrow$ Model 1
- Sample Dataset 2 $\rightarrow$ Slightly Different Model 2
- Sample Dataset 3 $\rightarrow$ Slightly Different Model 3

If the training sample data changes, the estimator also changes. So, $\hat{f}(x)$ is **random**. Different sample datasets produce different learned models.

So mathematically: $\hat{f}(x)$ is a _____ for a fixed input x .

In the equation $Y = f(x) + \varepsilon$, even for the same input X , the value of Y can be different because of random noise ε . Thus, Y can also be treated as a _____.

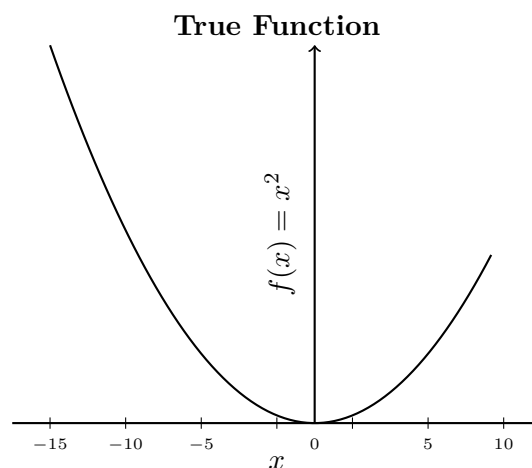
SECTION 3 Scenario: The God's-Eye View

▷ HOOK

Now consider yourself as a “God” who knows the true function of a population:

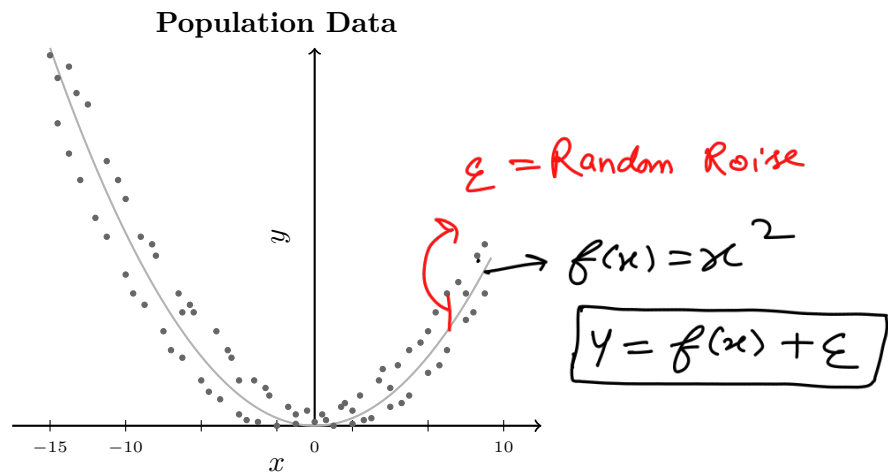
$$Y = f(x) = x^2 \quad \text{where } x \in [-15, 10]$$

Visual: True function $f(x) = x^2$



Now suppose we add random noise to this function to get 100 new data points. These become the **population data**.

Visual: Population data (true function + noise)



Now suppose we draw out **3 random sample sets** from this population data and give each to a different person. Each person builds an ML model.

3.1 Case 1: All three build a straight-line model (without knowing the true function)

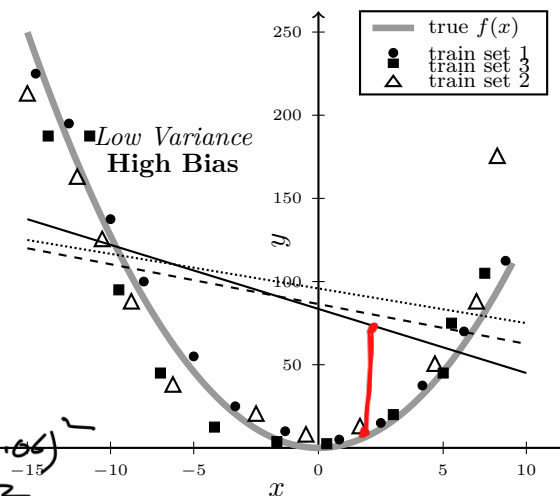
Simple Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x$$

(Low complexity)

$$\begin{aligned} \text{Bias}(\hat{f}(1)) &= E[\hat{f}(1)] - f(1) \\ &= \frac{(2 + 2.15 + 2.05)}{3} - 1 \end{aligned}$$

$$\text{Var}(\hat{f}(1)) = \frac{(2 - 2.06)^2 + (2.15 - 2.06)^2 + (2.05 - 2.06)^2}{3}$$



● → Sample 1 (—)

■ → Sample 2 (---)

△ → Sample 3 (.....)

$$\begin{aligned} \hat{y} &= 1.1 + 0.9x & \hat{y} &= 2.0 \quad x=1 \\ \hat{y} &= 1.2 + 0.95x & \hat{y} &= 2.15 \\ \hat{y} &= 1.05 + 1.0x & \hat{y} &= 2.05 \end{aligned}$$

$f(x) = x^2$
 $f(1) = 1$

$\bar{y} = 2.06$

All 3 datasets and their straight-line models are shown on a single graph. Notice that the three lines are very close to each other (low variance) but none of them can capture the parabolic shape (high bias).

All three linear models are **not able to fit** their respective training data properly.

These models have **High Bias**: These are unable to capture the true pattern of data.

These models have **Low Variance**: All 3 model equations / parameters are almost the same.

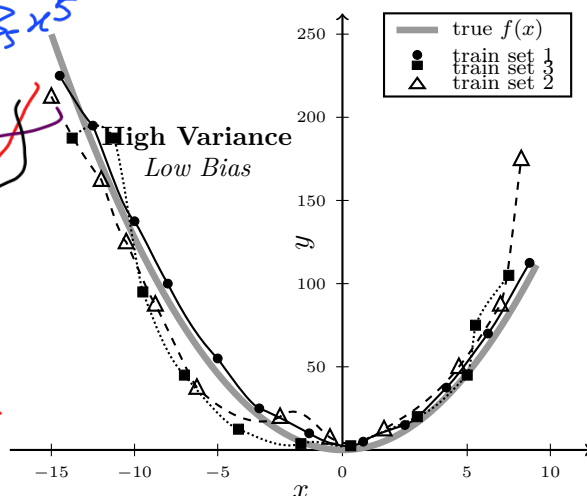
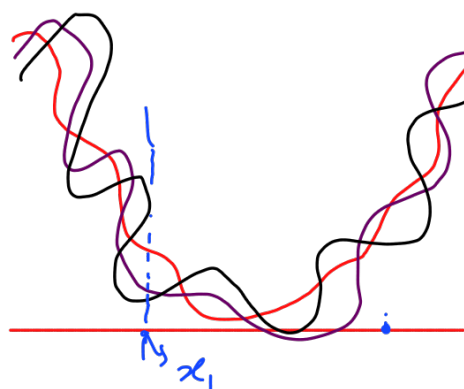
These 3 models are _____ models.

Underfitting

→ Highly complex (Polynomial Regression Model)

3.2 Case 2: All three increase the degree of their models

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_5 x^5$$



— (M1)
— (M2)
— (M3)

These models are very sensitive to sample datasets.

All 3 datasets and their high-degree models are shown on a single graph. Each curve passes through its own training points (low bias) but the three curves are wildly different from each other (high variance).

These models are trying to **exactly fit** their training datasets.

These models have **Low Bias**: Every prediction / curve is passing exactly from the actual data point.

These models have **High Variance**: The equation / parameters of the models are very diff.

These 3 models are **overfitting** models.

→ Prediction at each point is also very diff.

PRACTICE P1 · BIAS AND VARIANCE INTUITION

(a) Complete the table:

Model type	Bias	Variance	Fitting behaviour
Too simple (e.g. straight line for x^2)	_____	_____	_____
Too complex (e.g. degree-20 polynomial)	_____	_____	_____
Appropriately complex	_____	_____	_____

(b) **True or False:** If the training data changes, an underfitting model's predictions will change drastically. []

Justify: _____

(c) **True or False:** A model with high bias always has high variance. []

Justify: _____

SECTION 4 The Bias–Variance Tradeoff

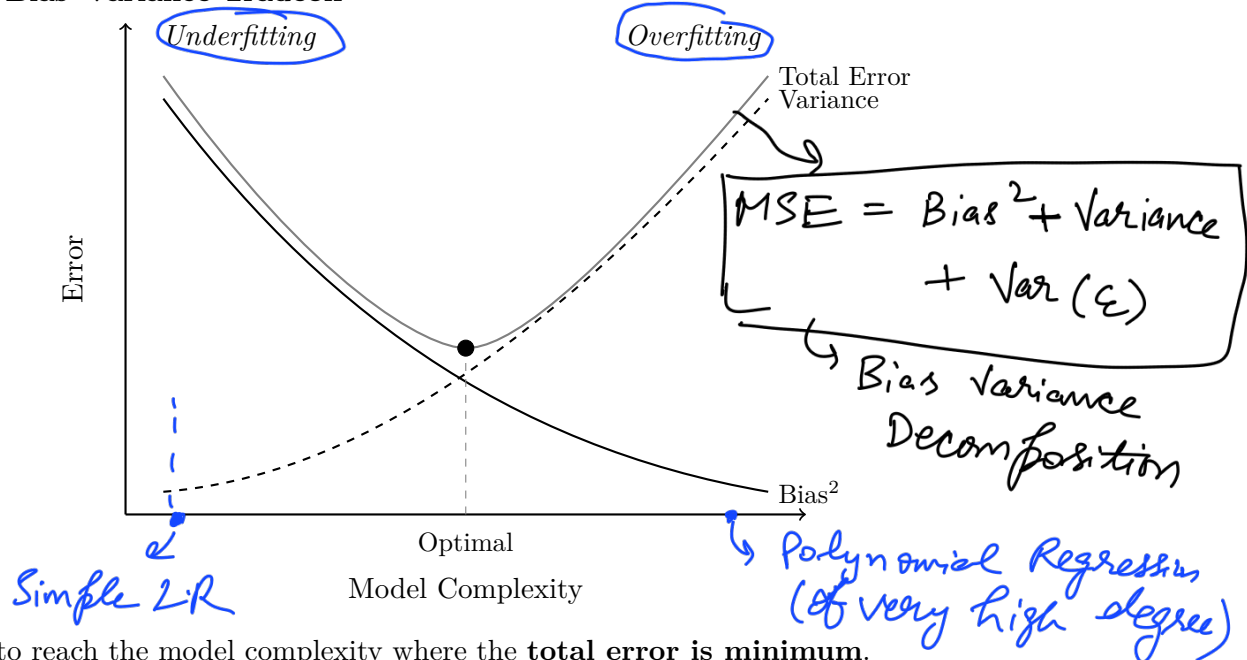
▷ HOOK

When you try to **decrease the bias** of a model, its variance automatically starts **increasing**. This is known as the **Bias–Variance Tradeoff**.

When Bias $\uparrow$, Variance $\downarrow$ and vice versa.

Both Bias and Variance are forms of errors. Their relationship with model complexity follows this graph:

Visual: Bias–Variance Tradeoff



We aim to reach the model complexity where the **total error is minimum**.

SECTION 5 Revisiting the Discussion: Mathematical Definitions

▷ HOOK

Let us revisit the equation:

$$Y = f(x) + \varepsilon$$

where $f(x)$ = true population function, Y = observed output, ε = random noise = irreducible error.

$$E(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2 = E(\varepsilon^2).$$

We have an estimator of the true function, $\hat{f}(x)$, which is also a random variable.

5.1 Bias

$$\text{Bias}(\hat{f}(x)) = E[\hat{f}(x)] - f(x)$$

where:

- $E[\hat{f}(x)]$ = Expected value of the prediction over _____
- $f(x)$ = _____

Unbiased Predictor:

$$\text{Bias}(\hat{f}(x)) = 0 \implies E[\hat{f}(x)] = \underline{\hat{f}(x)} \text{ for all } x.$$

Example: Say you have 5 models with slopes m_1, m_2, m_3, m_4, m_5 . If $E(m) = \gamma$ or $\frac{\sum m_i}{5} = \gamma$, where γ is the slope of the true function, then we can say the model is an _____.

$\hat{f}(x) \rightarrow$ Prediction of my model at any point x .
 $f(x) \rightarrow$ True Function (unknown)

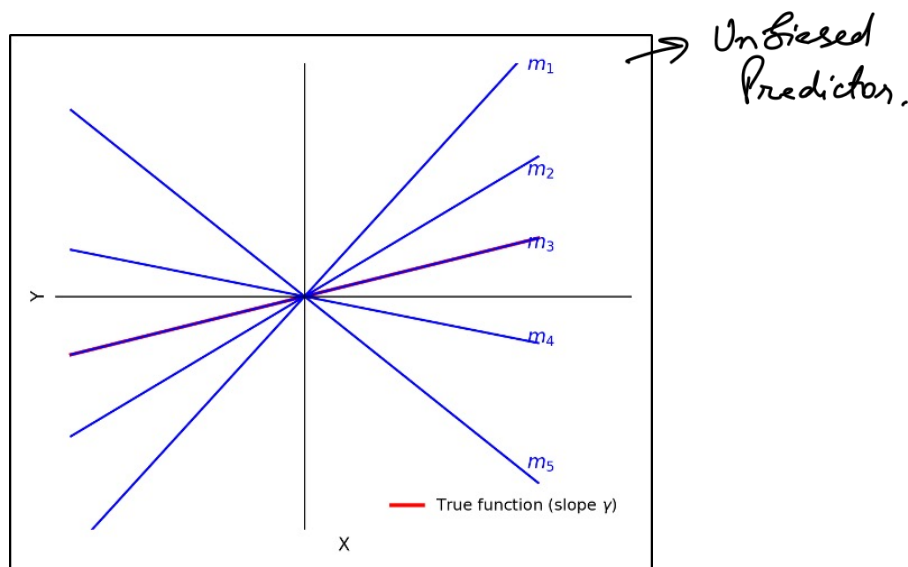


Fig. 4: Five models with different slopes (m_1 – m_5) and the true function (slope γ).

5.2 Variance

$$\text{Var}(X) = E[(X - E(X))^2]$$

Mathematically, we calculate the variance of our predictor:

$$\text{Var}(\hat{f}(x)) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

Interpretation:

- Train on many sample datasets.
- Observe prediction spread.

How much the output of different models trained on diff. sample datasets varying with each other.

SECTION 6 MSE and the Bias–Variance Decomposition

6.1 Mean Squared Error (MSE)

At a point x :

$$\text{MSE}(x) = E[(Y - \hat{f}(x))^2]$$

This measures the _____.

6.2 Bias–Variance Decomposition — Final Result

$$\text{MSE} = \text{Bias}^2 + \text{Variance} + \text{Var}(\varepsilon)$$

Remember

or equivalently:

$$\text{MSE} = \underbrace{\text{Bias}^2 + \text{Variance}}_{\text{Reducible Error}} + \underbrace{\text{Var}(\varepsilon)}_{\text{Irreducible Error}}$$

△ WARNING

The proof of the above Bias–Variance Decomposition result is beyond the scope of this lecture. Students who are interested in the proof are welcome to ask the instructor about it separately after the lecture.

$$\begin{aligned} \text{Bias}(\hat{f}(x)) &= E(\hat{f}(x)) - f(x) \\ \text{Var}(\hat{f}(x)) &= E[(\hat{f}(x) - E(\hat{f}(x)))^2] \end{aligned}$$

We can only Infer about Bias & Variance

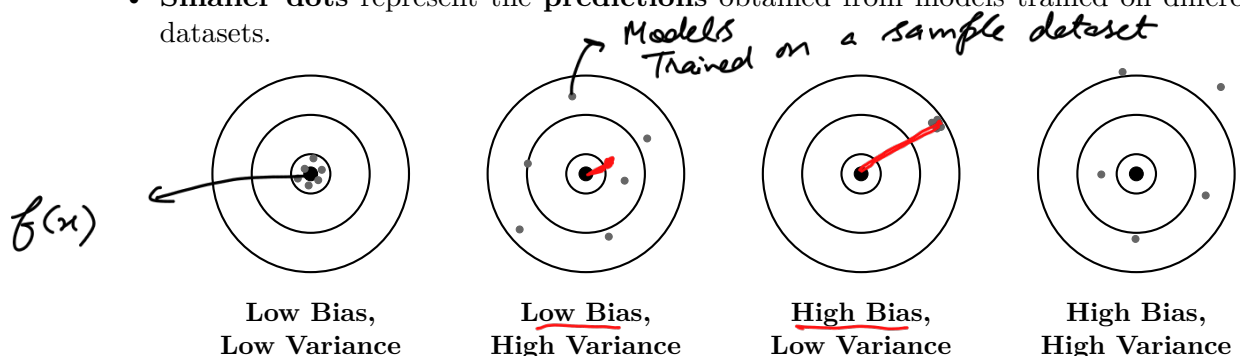
I need $\rightarrow f(x)$ (True function)
 $\rightarrow$ I can't draw Infinite samples
 $\rightarrow$ I can't calculate True Bias.

SECTION 7 Graphical Illustration of Bias and Variance

Visual: Dartboard analogy

Imagine throwing darts at a dartboard multiple times. Each throw represents training a model on a different sample dataset and making a prediction.

- **Concentric circles** represent increasing distance from the true value — the further out, the larger the error.
- **Bold dot at the centre** (bull's-eye) represents the **true value** we are trying to predict.
- **Smaller dots** represent the **predictions** obtained from models trained on different sample datasets.



PRACTICE P2 · MATHEMATICAL CONCEPTS

- (a) If $E[\hat{f}(x)] = 12$ and $f(x) = 10$ at a particular point x , what is the bias?
Bias = _____
- (b) If the bias is 2, the variance is 3, and $\sigma^2 = 1$, what is the MSE?
MSE = _____
- (c) **Match the Column.** Match each dartboard pattern with the correct characterisation.

	Dartboard Pattern	Characterisation
1	Dots tightly clustered around the bullseye	(A) High Bias, High Variance
2	Dots tightly clustered far from the bullseye	(B) Low Bias, Low Variance
3	Dots spread widely around the bullseye	(C) High Bias, Low Variance
4	Dots spread widely far from the bullseye	(D) Low Bias, High Variance

Answers: 1 → _____ 2 → _____ 3 → _____ 4 → _____

- (d) **True or False:** Irreducible error can be reduced by choosing a more complex model. []
Justify: _____
- (e) A model has Bias² = 1, Variance = 9, σ^2 = 2. What is the total error? What component dominates?

PART II: BRIDGING THEORY WITH PRACTICE

SECTION 8 The Mathematical World vs. The Real World

▷ HOOK

All this mathematics makes sense. But in real life, how do we actually calculate Bias and Variance? This is perhaps one of the biggest differences between learning Machine Learning **theoretically** and doing Machine Learning **in practice**.

Question 1: In real-world problems, do we have access to the true function $f(x)$? Why or why not?

Question 2: When you download a dataset (e.g. from Kaggle), does it represent the entire population? Explain.

So mathematically, we can think of it like this:

Unknown Population $\xrightarrow{\text{sample}}$ One Sample Dataset (e.g. Kaggle CSV) $\xrightarrow{\text{train}}$ Train One ML Model

★ KEY INSIGHT

We only have **one sample dataset**, not hundreds of different sample datasets like we imagined while defining Bias and Variance.

Therefore, the true mathematical Bias and Variance **cannot be calculated** in real-world ML problems. Their definitions require:

- The true function $f(x)$ — which is unknown.
- Predictions from many models trained on many different sample datasets — which we do not have.

The mathematical definitions of Bias and Variance are **theoretical definitions**. They help us understand how learning algorithms behave, but they usually cannot be computed exactly in practice.

Then why do we study them?

Because they explain:

- why models make errors
- why overfitting happens
- why underfitting happens
- why increasing model complexity is not always beneficial



Even though we cannot compute the exact values, the **concepts guide almost every important decision** in ML.

SECTION 9 Diagnosing High Bias and High Variance

▷ HOOK

Instead of trying to calculate the exact Bias and Variance, practitioners try to **infer** whether a model is suffering from high Bias or high Variance.

Think of a doctor. A doctor usually cannot directly measure pain. Instead, the doctor looks at **symptoms**. Similarly, an ML engineer looks at evaluation results to **diagnose** the model.

9.1 Diagnosing High Bias (Underfitting)

Training Accuracy = 70%
Validation Accuracy = 68%

} Underfitting
Bias High
Var Low

Both are _____. If the model cannot even learn the training data properly, it is probably too _____.

This is a sign of **High Bias (Underfitting)**.

9.2 Diagnosing High Variance (Overfitting)

Training Accuracy = 99%
Validation Accuracy = 82%

*Low Bias
High Var.*

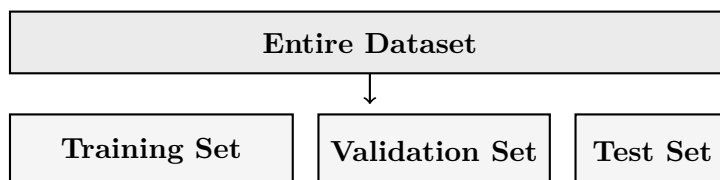
The model performs extremely well on training data but much _____ on unseen data. It has **memorized** the training data instead of learning general patterns.

This is a sign of **High Variance (Overfitting)**.

SECTION 10 Why Do We Split the Dataset?

► HOOK

In practice, we cannot collect many different datasets from the population. Instead, we try to **simulate the idea** by splitting our available dataset.



- The **training set** is used to _____.
- The **validation set** is used to check whether the model _____.

Diagnosing from splits:

- Model performs well only on training data, poorly on validation data $\Rightarrow$ suspect _____.
- Model performs poorly on both $\Rightarrow$ suspect _____.

Although this does not calculate Bias and Variance mathematically, it gives us **practical evidence** about the behaviour of the model.

SECTION 11 Cross-Validation

Suppose we split the dataset **only once**. The result may depend heavily on that particular split.

To reduce this dependence, we use **Cross-Validation**. Instead of training the model once, we train it **multiple times** using different train-validation splits. This allows us to evaluate how stable the model is across different subsets of the available data.

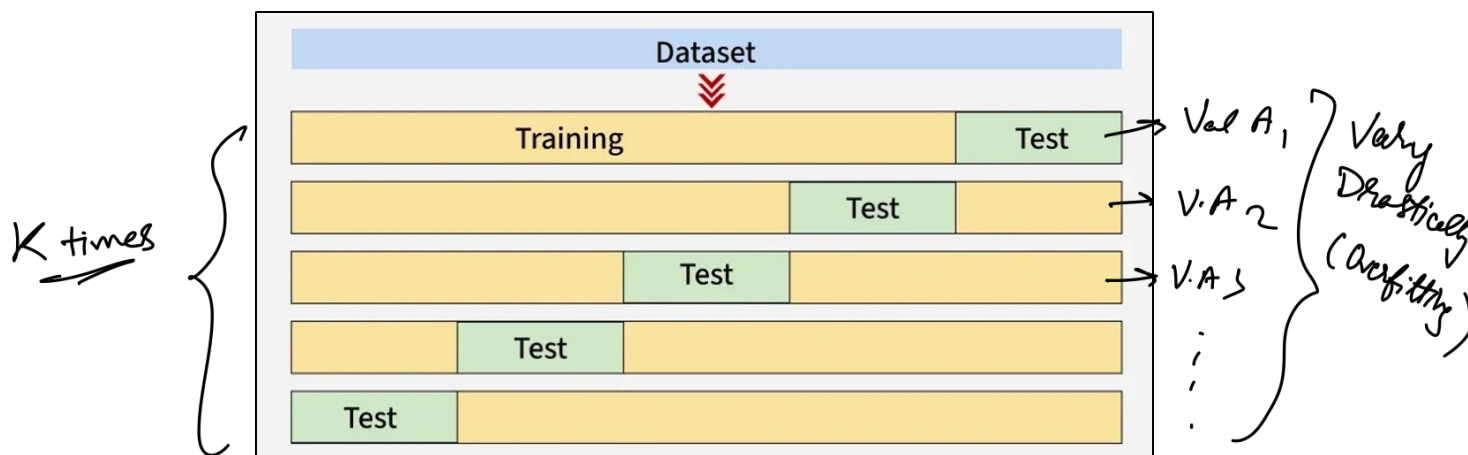


Fig. 5: K-Fold Cross-Validation — each fold takes a turn as the test set.

If the performance changes significantly across different folds, it suggests that the model is **sensitive to the training data** — a practical indication of _____.

◆ TAKEAWAY

You can study Cross-Validation in detail as **homework**. The key idea is: training the model multiple times on different splits helps us estimate how much the model's performance varies — which is closely related to the concept of Variance.

SECTION 12 Learning Curves Give Another Clue

▷ HOOK

A **learning curve** shows how the training and validation errors change as we increase the amount of training data. Without ever knowing the true function, these curves help us understand what kind of problem our model is facing.

Visual: Training Error and Validation Error vs. Model Complexity

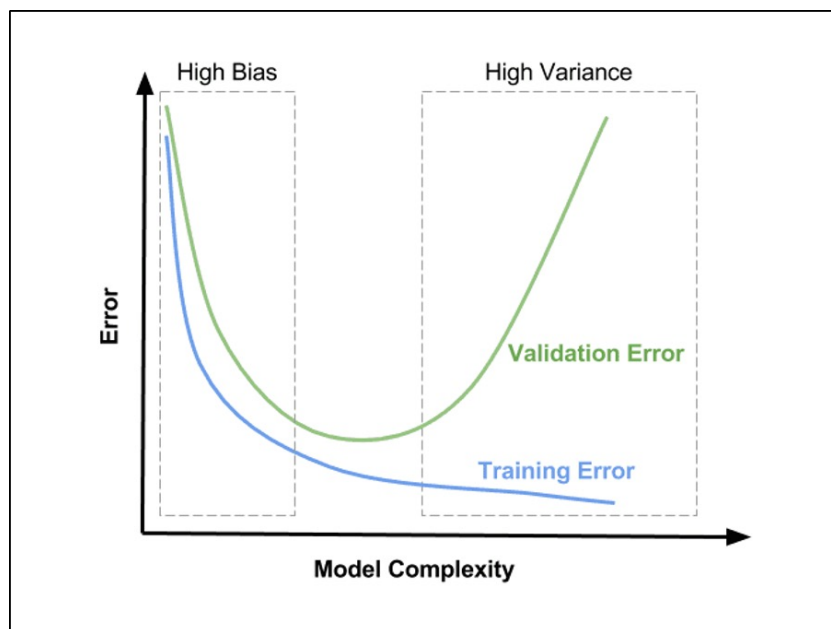


Fig. 6: How training and validation errors behave as model complexity increases.

Typical observations:

- **High Bias:** Both training and validation errors remain _____, even after adding more data.
- **High Variance:** Training error is very _____, but validation error is much _____. As more data is added, the validation error often _____.

PRACTICE P3 · DIAGNOSIS AND APPLICATION

- (a) For each scenario, diagnose whether the model suffers from **High Bias**, **High Variance**, or a **Good fit**.

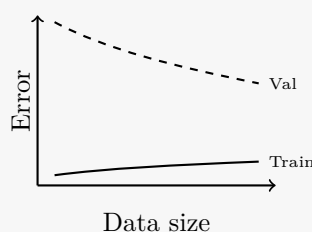
Scenario	Train Acc.	Val Acc.	Diagnosis
(i)	98%	97%	_____
(ii)	65%	63%	_____
(iii)	99%	72%	_____
(iv)	55%	54%	_____

- (b) A team trains a decision tree with no depth limit on a house price dataset. Training RMSE is very low but validation RMSE is 3 times higher.

(i) What problem is this? _____

(ii) Name **two** ways to fix it: _____ and _____

- (c) Look at the learning curve below and answer:



(i) Is this High Bias or High Variance? _____

(ii) Will adding more training data likely help? _____

(iii) Would increasing model complexity help? _____

- (d) **Assertion–Reason:**

A: A linear regression model applied to a highly non-linear dataset is likely to have high bias.

R: A linear model cannot capture non-linear patterns, so it will systematically underfit the data.

- (i) Both A and R are true, and R is the correct explanation of A. []
 (ii) Both A and R are true, but R is not the correct explanation of A. []
 (iii) A is true but R is false. []
 (iv) A is false but R is true. []

SECTION 13 Final Takeaway**▷ HOOK**

Let us summarise everything we have learned.

Mathematically:

- Bias and Variance are defined over all possible sample training datasets drawn from the same population.
- Their definitions assume that the true function $f(x)$ is known.
- Therefore, they are **theoretical quantities** that help us understand how learning algorithms behave.

Practically:

- We never know the true function.
- We usually have only one sample dataset.
- Therefore, we cannot calculate the true Bias and Variance directly.

Instead, we **infer** them using practical tools:

Practical Tool	What it reveals
Training & Validation Performance	_____
Train–Validation–Test Split	_____
Cross-Validation	_____
Learning Curves	_____

PRACTICE P4 · COMPREHENSIVE CHALLENGE*(Homework)*

(a) **Fill in the Key Terms table:**

Term	Definition
Estimator	_____
Bias	_____
Variance	_____
Underfitting	_____
Overfitting	_____
Irreducible Error	_____

(b) **True or False:**

Statement	T/F	Justification
The true Bias and Variance can always be calculated from a real-world dataset.	[]	_____
$MSE = \text{Bias} + \text{Variance} + \text{Irreducible Error}$.	[]	_____
A model with very high training accuracy but low validation accuracy likely has high variance.	[]	_____
Adding more data always fixes a high bias problem.	[]	_____
Cross-validation helps estimate the stability of model performance.	[]	_____

(c) **Find the Mistake.** A student says: “My model has 99% training accuracy and 98% validation accuracy. It must be suffering from high variance because the training accuracy is

very high.”

- (d) **Predict the Outcome.** A data scientist uses a degree-1 polynomial to model a dataset where the true relationship is $y = x^3 + 2x^2 - x + 5$. What will happen to the Bias and Variance? Will the model underfit or overfit?
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BY THE END OF THIS SHEET, YOU CAN

- ☐ Write the equation $Y = f(x) + \varepsilon$ and explain each term.
- ☐ Explain why $\hat{f}(x)$ is a random variable.
- ☐ Define Bias and Variance mathematically.
- ☐ State the Bias–Variance Decomposition: $\text{MSE} = \text{Bias}^2 + \text{Variance} + \text{Var}(\varepsilon)$.
- ☐ Use the dartboard analogy to visualise all four bias-variance combinations.
- ☐ Explain the Bias–Variance Tradeoff and its relationship to model complexity.
- ☐ Explain why true Bias and Variance cannot be computed in practice.
- ☐ Diagnose High Bias and High Variance from training/validation performance.
- ☐ Interpret learning curves to identify underfitting and overfitting.